LEARNER GUIDE · COURSE SLIDES

Designing and Implementing Microsoft DevOps Solutions (AZ-400)

WSQ Course Code: TGS-2024042605

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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TRAQOM + SSG DIGITAL ATTENDANCE
Digital Attendance (Mandatory)

- Take AM, PM and assessment digital attendance where required.
- Use the registered learner identity throughout the course.
- Minimum attendance requirements apply for funding and assessment records.

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YOUR TRAINER · GENERAL

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS
PhD — specialises in AI, automation and software engineering.

DELIVERS
WSQ courses on AI agents, automation (n8n), app development and DevOps delivery.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER
Let's Know Each Other

- Your name and organisation or role.
- Your experience with DevOps, Azure DevOps, GitHub or CI/CD.
- One delivery problem you want automation to solve.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL
LMS / TMS

- Access course materials, attendance and assessment through the LMS/TMS portal.
- Use the Learner Guide and copied lab source for step-by-step practice.

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GET THE LABS
Download the Course Labs

1. Open the AZ-400 lab repository from the course link.
2. Use the copy in courseware/labs-source for offline reference.
3. Complete the labs in order.
4. Keep screenshots, URLs or pipeline run links as evidence.

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SCHEDULE

Lesson Plan - 2 Days, 8 hours/day

Day 1: DevOps process, source control, YAML CI and release pipelines.
LP schedule includes slide numbers for each activity.

Day 2: IaC, DevSecOps, monitoring and capstone integration.

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WHAT YOU'LL ACHIEVE
Learning Outcomes

1. Design DevOps processes with traceability from work item to deployment.
2. Implement source control, pull requests and review practices.
3. Create YAML CI pipelines with tests and artifacts.
4. Apply IaC, DevSecOps, monitoring and feedback loops.
5. Complete an end-to-end AZ-400 capstone workflow.

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BRIEFING FOR ASSESSMENT

Assessment Briefing

- Follow trainer instructions for written and practical assessment evidence.
- No unauthorised materials, photos or recording during assessment.
- Submit required evidence through the assigned platform.

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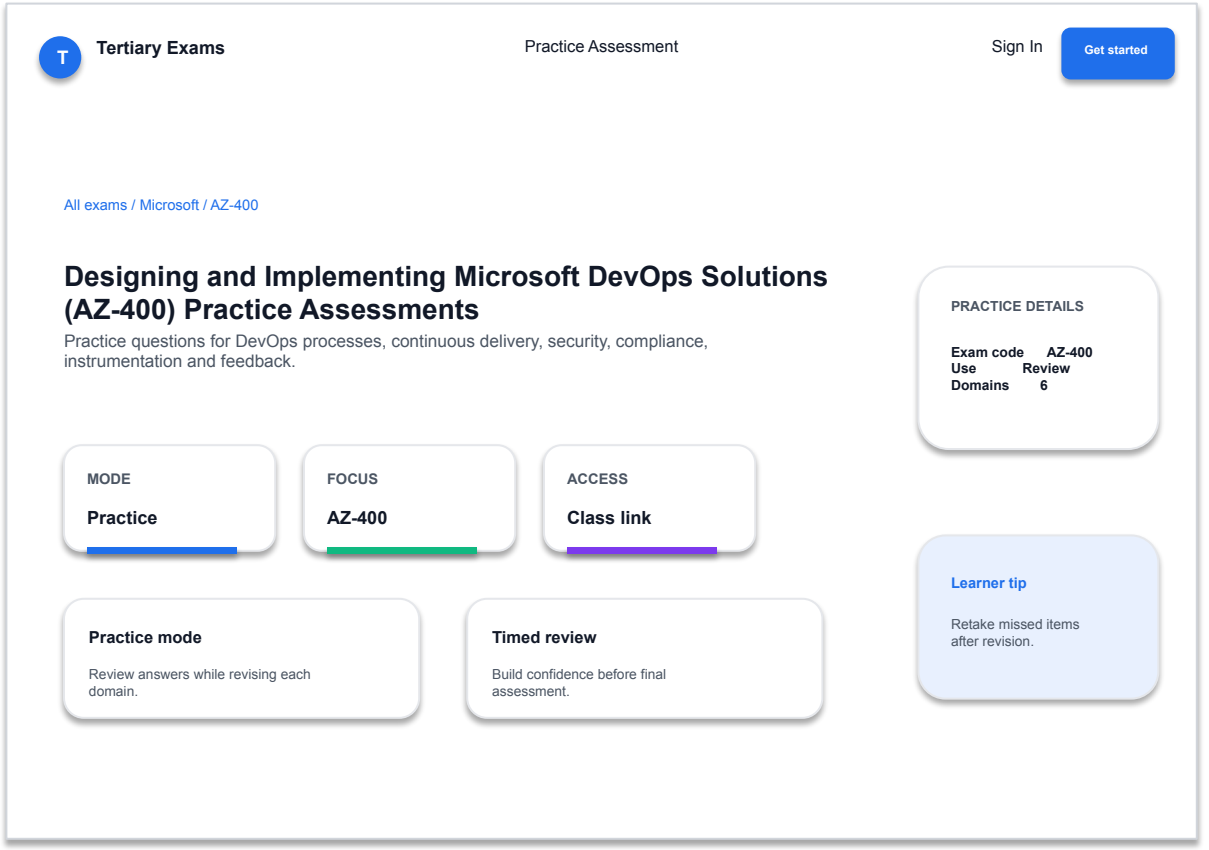
PRACTICE ASSESSMENT

Take the Online AZ-400 Practice Assessments

- Reinforce the AZ-400 delivery lifecycle with realistic, scenario-style questions before assessment day.
- Use practice sets to test DevOps processes, source control, CI/CD, IaC, security and monitoring.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete practice questions from Google Classroom, then bring difficult items to the review session.

Access and practise at:

Google Classroom > Practice Assessments



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ASSESSMENT

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

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Azure DevOps Delivery Lifecycle

Plan + Code + Build + Release + Operate

The eight labs build one traceable DevOps solution in sequence.

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OVERVIEW
AZ-400 is about designing delivery systems

- The course focuses on repeatable software delivery, security, governance and feedback.
- Learners practise design tradeoffs, not only tool commands.

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FEATURES

Core DevOps Capabilities Practised

Boards and traceability

IaC, DevSecOps, monitoring and feedback

Repos, branches and pull requests

YAML CI, artifacts and releases

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WHY DEVOPS

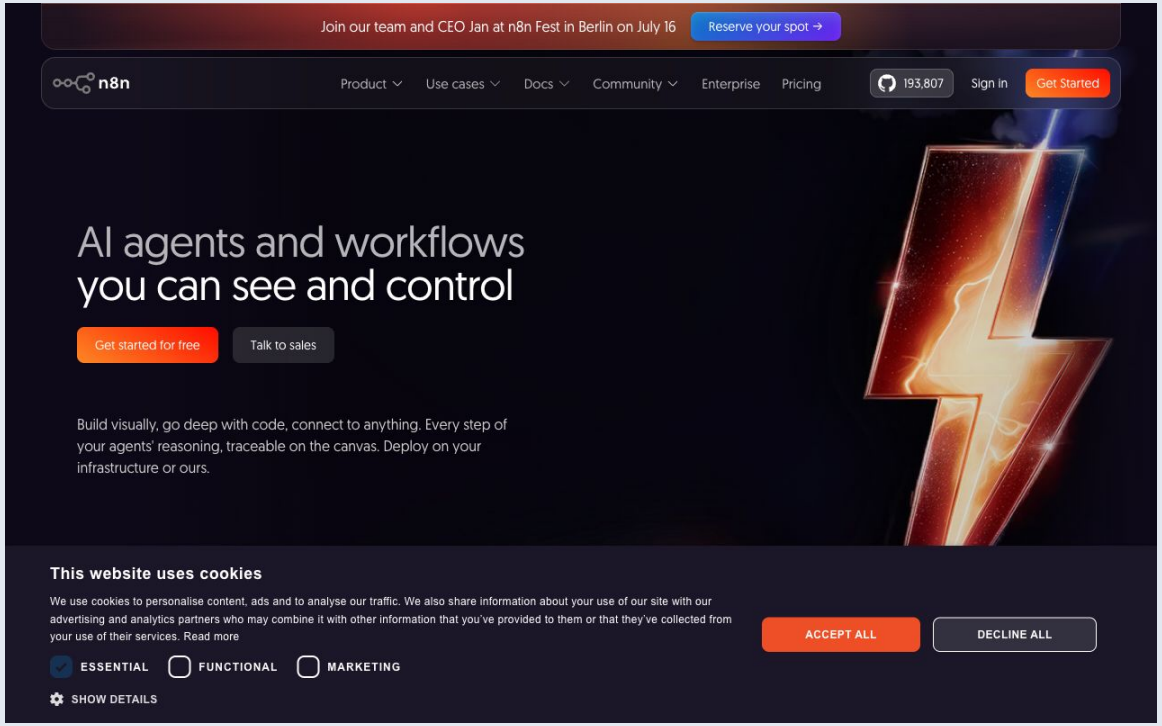
DevOps connects work, code and operations

- Traceability shows why a change exists, how it was built, where it deployed and what happened after release.
- This is the thread followed through all labs.

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THE PLATFORM
Azure DevOps and GitHub

- Use Azure DevOps or GitHub for work tracking, repositories, pipelines and delivery evidence.
- Use Azure, Bicep and monitoring tools for deployment and operations practice.



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MOTIVATION
Why automate delivery?

- Reduce manual release risk.
- Make builds, tests and deployments repeatable.
- Create fast feedback from code, security, telemetry and users.

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Lab 01 - DevOps Processes, Boards, and Traceability

LAB 01

Create a work tracking structure. Link work to code and releases. Define DevOps metrics.

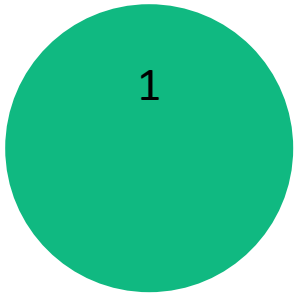
You'll build

Boards → Branch → Commit → Pull Request → Dashboard

Key tools: Azure DevOps

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STEP 1 OF 10



Create or open an Azure DevOps project or GitHub repository.

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✅ Test it

- Work items follow a clear hierarchy.
- Traceability includes work, code, pipeline, and release evidence.
- Metrics are defined.
- Documentation is readable.

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Lab 02 - Source Control, Branching, and Pull Requests

LAB 02

Choose a branch strategy. Configure pull request workflow. Apply branch protection or policies.

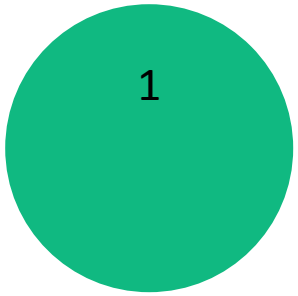
You'll build

Repository → Branch → Pull Request → Review

Key tools: Azure DevOps, Git, Repos

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STEP 1 OF 11



Create or open a Git repository.

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✅ Test it

- Branch strategy fits the team scenario.
- Pull request includes review and checks.
- Tagging convention is documented.
- Git recovery step is successful.

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Lab 03 - YAML CI Pipelines, Testing, and Artifacts

LAB 03

Create a YAML CI pipeline. Configure triggers. Run tests.

You'll build

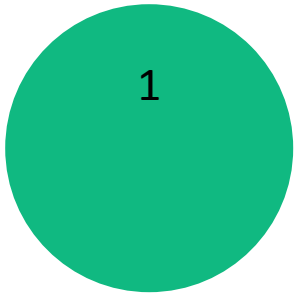
YAML Pipeline → Build → Test → Artifact

Key tools: Azure DevOps, Pipelines, Artifacts

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Lab 03 - YAML CI Pipelines, Testing, and Artifacts

STEP 1 OF 12



Use a sample application repository.

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✅ Test it

- Pipeline triggers correctly.
- Tests run.
- Artifact is published.
- Failure is documented and fixed.

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Lab 04 - Release Pipelines and Deployment Strategies

LAB 04

Design release stages. Configure environments and approvals. Compare deployment strategies.

You'll build

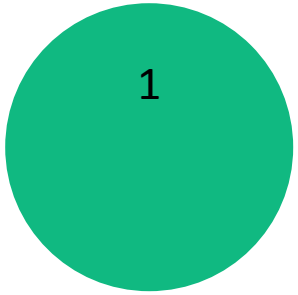
Artifact → Environment → Approval → Deployment

Key tools: Azure DevOps, Pipelines, Artifacts, Environments, Approvals

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Lab 04 - Release Pipelines and Deployment Strategies

STEP 1 OF 10



Define dev, test, staging, and production environments.

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✅ Test it

- Deployment strategy matches risk and downtime needs.
- Approval/check evidence is captured.
- Rollback path is clear.
- Release notes are generated or drafted.

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Lab 05 - Infrastructure as Code with Bicep and Templates

LAB 05

Define IaC strategy. Create a Bicep or template file. Validate deployment.

You'll build

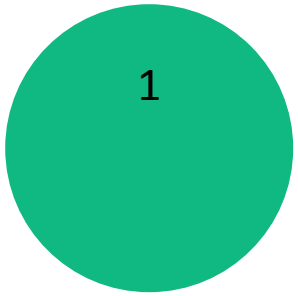
Bicep Template → Validation → Deployment

Key tools: Azure DevOps, Azure CLI, Bicep

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Lab 05 - Infrastructure as Code with Bicep and Templates

STEP 1 OF 10



Identify infrastructure required by the sample app.

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✅ Test it

- Template is parameterized.
- Tags are included.
- Validation or deployment evidence is captured.
- Drift and review controls are documented.

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Lab 06 - DevSecOps, Secrets, and Compliance Scanning

LAB 06

Manage secrets safely. Design secure pipeline authentication. Configure scanning checks.

You'll build

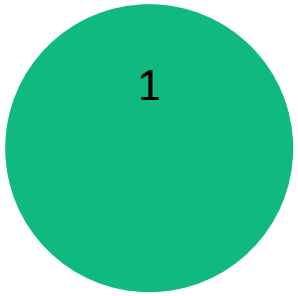
Secrets → Scanning → Compliance Gate

Key tools: Azure DevOps, Secrets, Security scanning

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Lab 06 - DevSecOps, Secrets, and Compliance Scanning

STEP 1 OF 10



Identify secrets used by the sample app or pipeline.

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✅ Test it

- No secrets are stored in code.
- Authentication design follows least privilege.
- Scanning plan covers code, dependencies, secrets, and containers.
- Evidence table is audit-friendly.

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Lab 07 - Instrumentation, Monitoring, KQL, and Feedback

LAB 07

Design monitoring strategy. Review Azure Monitor and Application Insights. Write KQL queries.

You'll build

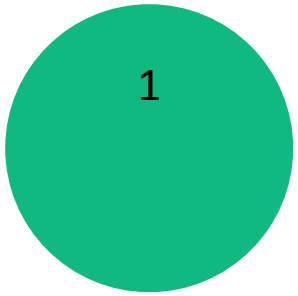
Telemetry → KQL → Dashboard → Feedback

Key tools: Azure DevOps, Application Insights, KQL

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Lab 07 - Instrumentation, Monitoring, KQL, and Feedback

STEP 1 OF 10



Identify telemetry required for the sample application.

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✅ Test it

- Metrics connect to operational decisions.
- Alerts have clear action.
- KQL queries are documented.
- Feedback loop is practical.

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Lab 08 - AZ-400 Capstone: End-to-End DevOps Solution

LAB 08

Design a complete DevOps solution. Combine work tracking, source control, pipelines, security, and monitoring. Make tradeoff decisions.

You'll build

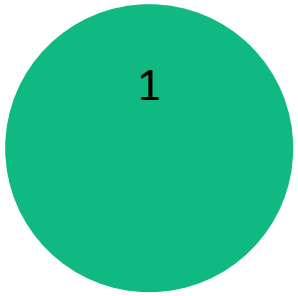
Plan → Code → Build → Release → Monitor

Key tools: Azure DevOps

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Lab 08 - AZ-400 Capstone: End-to-End DevOps Solution

STEP 1 OF 13



Review the trainer-provided scenario.

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✅ Test it

- Solution includes traceability.
- Pipeline strategy covers build, test, release, and rollback.
- Security and monitoring are built in.
- Study plan targets weak domains.

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WRAP-UP
Presentation & Assessment

- Present the problem, your design, evidence and what you would improve.
- Complete written and practical assessment tasks as instructed.
- Submit evidence through the LMS/TMS portal.
- Remember the assessment digital attendance requirement.

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